Abhay Kumar Singh **Senior Software Engineer**
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PROFILE

Senior Software Engineer with 9+ years of experience designing and building high-scale, distributed backend systems using Java, Spring Boot, Kafka, and cloud-native technologies. Experienced in real-time fraud, risk, and financial platforms processing high-volume transactional workloads. Strong expertise in distributed systems, concurrency, performance optimization, reliability engineering, and system design, with a track record of reducing latency and improving production reliability.

TECHNICAL SKILLS

- **Languages:** Java, Python, C++
- **Backend:** Spring Boot, REST, Kafka, Microservices, Spring Security, Hibernate, OAuth2, OIDC
- **Architecture:** System Design, Distributed Systems, Concurrency, Event-Driven Architecture, Resilience Patterns
- **Cloud & Infrastructure:** AWS, GCP, Docker, Kubernetes, Jenkins, Github Actions
- **Databases:** Cassandra, MySQL, MongoDB, Redis
- **Observability:** Splunk, Dynatrace, Kibana

EXPERIENCE

Application Developer, IBM Corp | Phoenix, AZ | July 2025 – Present
- Built a real-time fraud decision engine for American Express that evaluates bank funding account eligibility, reducing decision latency by 20% through parallel downstream integration and asynchronous processing, supporting 12+ banking products.
- Played a key role in launching digital banking capabilities by enabling secure account linking at scale and enabling adoption of new banking products.
- Orchestrated parallel integrations with downstream data providers (Plaid, EWS, Yodlee etc.) using a two-wave execution strategy to optimize latency while maximizing data enrichment.
- Implemented distributed observability across 3 banking services using Splunk, Dynatrace, and Kibana, reducing production diagnosis time by 40% and improving incident detection.
- Authored a reusable GitHub Actions workflow to automate rolling deployments, enabling zero-downtime releases.
- Improved API response time by 50% through request optimization and asynchronous processing.

Software Engineer, IntraEdge Inc | Phoenix, AZ | Oct 2023 – July 2025
- Designed and developed scalable microservices for enterprise Risk Decision Engine handling credit risk, fraud detection, and eligibility decisions at American Express.
- Built high-throughput REST APIs processing with P99 latency below 300ms for real-time risk decisions.
- Designed and implemented a Redis caching layer for high throughput customer-facing APIs improving P95 latency by 65% and reducing downstream database load by ~55%.
- Led production deployments and resolved critical issues in a high-availability environment.
- Contributed to low-level system design and architecture reviews.

Software Engineer Lead, Capgemini America Inc | Phoenix, AZ | Nov 2022 – Jul 2023
- Led service scalability enhancements in performance-sensitive systems.
- Solved capacity and production stability challenges.
- Mentored engineers on debugging and architecture best practices.

Technology Lead, Infosys Limited | Phoenix, AZ | Sep 2018 – Nov 2022
- Delivered backend systems supporting subscription services for millions of credit-card users.
- Designed APIs and services across multiple projects.
- Contributed to low-level design and architectural reviews.
- Led teams through release cycles and production deployments.

Software Engineer, Innodata Inc | Noida, India | Jan 2017 – Aug 2018
- Developed enterprise workflow applications.
- Designed service-layer architecture.
- Integrated backend APIs with frontend systems.
  Employment: Innodata Inc. (Jan, 2017 – Aug, 2018), Outworks Solution Pvt. Ltd. (Dec, 2015 – Dec, 2016)

EDUCATION

- **Executive MBA,** Indian Institute of Management Kashipur, India, 2017.
- **B. Tech. Computer Science & Engineering,** Gurukula Kangri University, Haridwar India, 2012.